

Recovering Risk Neutral Densities from Option Prices: A New Approach

by

Leonidas S. Rompolis^(a) and Elias Tzavalis^(b)

Abstract

In this paper we present a new method of approximating the risk neutral density (RND) from option prices based on the C-type Gram-Charlier series expansion (GCSE) of a probability density function. The exponential form of this type of GCSE guarantees that it will always give positive values of the risk neutral probabilities and it can allow for stronger deviations from normality, which are two drawbacks of the A-type GCSE used in practice. To evaluate the performance of the suggested expansion of the RND, the paper presents simulation and empirical evidence.

KEYWORDS: Risk neutral density, Risk neutral moments, Gram-Charlier series expansion.

JEL: C10, C13, G13

The authors would like to thank an anonymous referee and the editor Paul Malatesta for their comments and suggestions on an earlier version of the paper.

^(a)Department of Public and Business Administration, University of Cyprus. 9-11 Larnakos str, Nicosia, Cyprus. *Email:* rompolis@ucy.ac.cy. Leonidas S. Rompolis acknowledges financial support from the Iraklitos Fellowships of Research of the Athens University of Economics and Business.

^(b) Department of Economics, Athens University of Economics and Business, 76 Patission str, Athens 104 34, Greece. *Email:* e.tzavalis@aueb.gr

I. Introduction

The efficient estimation of the conditional risk neutral density (RND) of the future price of an asset (e.g. stock), or its log-return, based on European option prices is of fundamental interest in the financial literature.¹ This density is used by traders and/or policy makers to assess market beliefs about future movements in the price of the underlying asset. To recover this density, existing methods in the literature rely on closed form solutions of it, or its characteristic function, provided by affine parametric option pricing models (see Heston (1993) and Bakshi, Cao and Chen (1997)) or they are based on nonparametric density estimation methods (see Ait-Sahalia and Lo (1998)). The former of these methods require to specify the price and volatility dynamics of the underlying asset which may not be always consistent with the data in hand. Note that for wider class of these models, such as the non-linear stochastic volatility model (see Ait-Sahalia (2002)), which generate a more realistic time evolution of the asset prices and market volatility, there is no closed-form density or characteristic function solution (see Bakshi, Ju and Ou-Yang (2006)). On the other hand, the nonparametric methods require large cross-sectional samples of option prices data (see Pagan and Ullah (1999)), which are not often available in practice.

Another approach frequently used in practice to retrieve the RND is based on density approximation by employing the standard Gram-Charlier series expansion (GCSE) around the normal, known as A-type (A-GCSE) (see Madan and Milne (1994), Corrado and Su (1996a,b), and Knight and Satchell (2001)). This expansion has been suggested in the literature by Jarrow and Rudd (1982) to account for non-normal distributions of stock log-returns, which allow for skewness and excess kurtosis. Although the A-GCSE can be proved to be a very handy and quick method of retrieving the RND even for small sets of cross-sectional option prices data, it has some drawbacks. First, it can provide negative probability values when the degree of skewness or kurtosis of the true density lie outside the following intervals $[-1.0493, 1.0493]$ and $[3, 7]$, respectively (see Jondeau and Rockinger (2001)). To avoid this problem, many authors have suggested that imposing a nonnegative constraint in the estimation of the RND approximated through the A-GCSE (see Jondeau and Rockinger (2001), and Flamouris and Giamouridis (2002)). However, this constraint

¹See Rubinstein (1994), Jackwerth and Rubinstein (1996), Melick and Thomas (1997), Ait-Sahalia and Lo (1998), Bliss and Panigirtzoglou (2002) and Abadir and Rockinger (2003).

can lead to serious biases in the estimates of the risk neutral probabilities. Second, as it relies on an expansion around the normal density, the A-GCSE can not account for stronger deviations from the RND implied by the Black and Scholes (BS) model. Finally, it may not satisfy the martingale restriction for the underlying asset (see Longstaff (1995)).

To encompass the above drawbacks of the A-GCSE, in this paper we propose another type of GCSE to approximate the RND, known as the C-type (C-GCSE) (see Charlier (1928)). This expansion is of an exponential form which always guarantees positive probabilities, for any degree of skewness and kurtosis of the true RND. In addition to this, it does not rely on any reference density around which to approximate the true RND. Implementation of the C-GCSE to recover the true RND from option prices only requires estimates of the moments of this density. These moments can be retrieved from option prices in a model-free manner following Bakshi and Madan's (2000) methodology, which relies on no-arbitrage conditions. Thus, the suggested type of GCSE can be thought of as providing a feasible approximation of the RND, without having to specify the underlying theoretical structure. The paper evaluates the efficiency of this type of GCSE to approximate the true RND by conducting two exercises. The first is based on a Monte Carlo simulation study, while the second empirically assesses its capacity to reduce option pricing valuation errors.

The paper is organized as follows. Section 2 presents the C-GCSE of the RND and discuss its properties in more detail. Sections 3 and 4 present the results of the simulation and empirical exercises, respectively, while Section 5 concludes the paper.

II. The C-type GCSE

Let r be the instantaneous riskless interest rate, $C_t(S_t, K)$ and $P_t(S_t, K)$ denote the price of a European call and put at time t , respectively, with maturity date T , where S_t stands for the current-time t price of the underlying stock and K is the strike price. Define the $(T-t)$ -period log-return of the underlying stock as $X_T(t) = \ln(\frac{S_T}{S_t})$ (or for notational simplicity as X). Then, the C-type Gram-Charlier series expansion (C-GCSE) considers that the RND of X can be written as the following infinite sum of Hermite polynomials

$$(1) \quad f_X(x) = \mathcal{C} \exp \left[\sum_{n=1}^{\infty} \frac{1}{n} \delta_n H_n(x^*) \right],$$

where $x^* = \frac{x - \mu_{X,1}}{\sigma_X}$ denotes the standardized values of X from its mean $\mu_{X,1}$, $H_n(\cdot)$ denotes the n^{th} -order Hermite polynomial, δ_n is the n^{th} -order series coefficient of the C-GCSE expansion and $\mathcal{C}$ is a normalizing constant scaling the values of $f_X(x)$ to sum up to unity. The Hermite polynomial $H_n(x^*)$ is defined as $H_n(x^*) = (-1)^n \frac{1}{\eta(x^*)} \frac{\partial^n \eta(x^*)}{\partial x^{*n}}$, where $\eta(\cdot)$ is the standard normal density. A natural choice of $\mathcal{C}$ is given by $\mathcal{C} = \left\{ \int \exp \left[\sum_{n=1}^{\infty} \frac{1}{n} \delta_n H_n(x^*) \right] dx \right\}^{-1}$.

Apart from always providing positive values of the risk neutral probabilities, the C-GCSE of the RND given by (1) has another interesting property. It does not require any reference density around which to expand the true RND, such as the normal implied by the BS model which is assumed by the A-GCSE.² This feature of the C-GCSE may permit a more accurate approximation of RNDs with stronger deviations from the normal density considered by the BS model than that based on the A-GCSE. In particular, it can also consider the first two moments of the RND, i.e. the mean and variance, to differ from the BS' model normal density. Ignoring such differences will lead to biased estimates of the true risk neutral probabilities over their whole interval support, as the approximating RND will not be even correctly located.³

²The fourth order A-GCSE of the RND is given by

$$f_X(x) = N(x) \left[1 + \frac{E_{X^*,3}}{3!} H_3(x^*) + \frac{E_{X^*,4}}{4!} H_4(x^*) \right]$$

where $N(x)$ (reference density) is the normal probability density function implied by the BS model, with conditional mean and variance of the log-return X given by $\mu_{X,1} \equiv E_t(X) = (r - \frac{1}{2}\sigma^2)(T-t)$ and $\sigma_X^2 \equiv Var_t(X) = \sigma^2(T-t)$, respectively. In the above expansion, $E_{X^*,n}$ denote the Hermite moments of X^* defined as the means of the Hermite polynomials, i.e.

$$E_{X^*,n} = \int H_n(x^*) f_{X^*}(x^*) dx^* = \int H_n(x^*) f_X(x) dx.$$

Note that for $n = 3, 4$, $E_{X^*,n}$ become the coefficients of skewness and excess kurtosis of X , respectively.

³To see this point more clearly, consider that option prices are driven by the stochastic volatility (SV) model. Then, the risk neutral conditional mean and variance of the log-return are given by the following formulas

$$E_t[X] = r(T-t) - \frac{1}{2} E_t \left[\int_t^T V_u du \right]$$

and

$$Var_t[X] = E_t \left[\int_t^T V_u du \right] + \frac{1}{4} Var_t \left[\int_t^T V_u du \right] - E_t \left[\int_t^T V_u du \int_t^T \sqrt{V_u} dW_u \right],$$

respectively, where $(V_t)_{t \in [0,T]}$ denotes the stochastic volatility process of the underlying stock price and $(W_t)_{t \in [0,T]}$ is a Brownian motion process that drives the stock price.

Substituting the above formula of the variance into that for the risk neutral mean of the reference density $N(x)$ employed in the A-GCSE, given as $E_t[X] = r(T-t) - \frac{1}{2} Var_t(X)$, implies that the conditional mean of the approximating RND $f_X(x)$ becomes

$$E_t[X] = r(T-t) - \frac{1}{2} E_t \left[\int_t^T V_u du \right] - \frac{1}{8} Var_t \left[\int_t^T V_u du \right] + \frac{1}{2} E_t \left[\int_t^T V_u du \int_t^T \sqrt{V_u} dW_u \right].$$

Truncating the C-GCSE of the RND $f_X(x)$, given by (1), at an optimal order (say m) can give an approximating density function whose values can be calculated based on option prices once risk neutral estimates of the series expansion coefficients δ_n are provided. To choose m , we can rely on one of the well known information criteria, such as Akaike's (AIC) and/or Schwarz (SC). These can be calculated based on the sum of squared errors between observed prices of options with different strike prices and their predicted values implied by the values of (1) retrieved from option prices, for any m .⁴ To obtain estimates of the series expansion coefficients δ_n , we can rely on the recently proposed method of Muscolino and Ricciardi (1999). According to this, the vector of the series expansion coefficients, denoted as $\boldsymbol{\delta}^{(m)} = (\delta_1, \delta_2, \dots, \delta_m)'$ for a given m , can be derived by solving out the following system of equations which can be written in matrix form as⁵

$$(2) \quad \mathbf{Y}^{(m)} \boldsymbol{\delta}^{(m)} = \mathbf{y}^{(m)},$$

where $\mathbf{y}^{(m)}$ is a m -dimension column vector, with elements functions of $E_{X^*,n}$

$$y_1^{(m)} = 0 \text{ and } y_n^{(m)} = -(n-1)E_{X^*,n-2}, \text{ for } n = 2, 3, \dots, m,$$

Note that the last two terms that appear in the last relationship are due to the stochastic nature of volatility. These terms appear under the A-GCSE of the RND which can lead to significant source of bias when estimating the mean of the true RND at any point of time t . The size of this bias depends on the persistency of the volatility of the SV model.

⁴According to these criteria, m is chosen to minimize the following quantity

$$\log(SSE_m/N) + (m+1)C(N)/N,$$

where SSE_m is the sum of squared option pricing errors for a given value of m , N is the number of available option prices and $C(N) = 2$ for the AIC and $C(N) = \log(N)$ for the SC (see Greene (2003)). The predicted values of option prices implied by the estimates of the RND can be obtained using the Cox and Ross (1976) option pricing valuation formula.

⁵In contrast to the A-GCSE, note that these expansion series coefficients are not directly related to the Hermite moments $E_{X^*,n}$. Due to the non-linear (exponential) nature of the C-GCSE, system (2) implies that the calculation of the m -dimension vector $\boldsymbol{\delta}^{(m)}$ requires knowledge of the first $(2m-2)$ Hermite moments. For instance, when $m = 4$ one should know the first six Hermite moments. Only in the simple case of $m = 2$, the solution of (2) requires knowledge of the first two Hermite moments. In this case, $\mathbf{Y}^{(2)} = \mathbf{I}_2$ and $\mathbf{y}^{(2)} = (0, -1)'$, and solving out (2) for $\boldsymbol{\delta}^{(2)}$ gives $\boldsymbol{\delta}^{(2)} = (0, -1)'$. By substituting the latter into (1), the RND becomes the normal one implied by the BS model.

and $\mathbf{Y}^{(m)}$ is a symmetric matrix of dimension $(m \times m)$, with elements

$$Y_{ij}^{(m)} = \sum_{n=0}^{i+j-2} \frac{1}{n!} \Delta_{i-1,j-1,n} E_{X^*,n},$$

where $\Delta_{p,q,r}$ are defined as follows

$$\Delta_{p,q,r} = \begin{cases} \prod_{a=p,q,r} \frac{\Gamma(a+1)}{\Gamma(b-a+1)} & \text{if } p+q+r = \text{even, } b = (p+q+r)/2 \geq p, q, r \\ 0 & \text{elsewhere,} \end{cases}$$

with $\Gamma(\cdot)$ denoting the Gamma function.

As system (2) indicates, to obtain risk neutral estimates of δ_n we need to derive estimates of the Hermite moments $E_{X^*,n}$, involved in arrays $\mathbf{Y}^{(m)}$ and $\mathbf{y}^{(m)}$, from option prices. These can be obtained by employing the following formula

$$(3) \quad E_{X^*,n} = \sum_{l=0}^{[n/2]} (-1)^l \frac{n^{[2l]}}{2^l l!} \mu_{X^*,n-2l},$$

where $[n/2]$ is the integer part of $n/2$, and $n^{[2l]} = \frac{n!}{(n-2l)!}$ (see Kendall and Stuart (1969)) and $\mu_{X^*,n}$ denotes the non-central moments of the standardized random variable X^* of order n given as

$$(4) \quad \mu_{X^*,n} = \frac{1}{\sigma_X^n} \sum_{h=0}^n \binom{n}{h} (-\mu_{X,1})^h \mu_{X,n-h},$$

where $\mu_{X,n}$ denotes the non-central moments of X , with $\sigma_X = \sqrt{\mu_{X,2} - \mu_{X,1}^2}$. Estimates of $\mu_{X,n}$ can be directly obtained from out-of-the-money (OTM) European call and put prices employing the formulas of the risk neutral moments (RNM) suggested by Bakshi, Kapadia and Madan (2003) for $n = 1, 2, 3, 4$ and, recently, extended by Rompolis and Tzavalis (2004)

to any order n as

$$\begin{aligned}\mu_{X,1} &= e^{r(T-t)} \left(1 - \int_{S_t}^{+\infty} \frac{1}{K^2} C_t(S_t, K) dK - \int_0^{S_t} \frac{1}{K^2} P_t(S_t, K) dK \right) - 1, \text{ for } n = 1, \\ (5) \mu_{X,n} &= e^{r(T-t)} \left\{ \int_{S_t}^{+\infty} \frac{n}{K^2} \left[\ln \left(\frac{K}{S_t} \right) \right]^{n-2} \left[n - 1 - \ln \left(\frac{K}{S_t} \right) \right] C_t(S_t, K) dK \right. \\ &\quad \left. + \int_0^{S_t} \frac{n}{K^2} \left[\ln \left(\frac{K}{S_t} \right) \right]^{n-2} \left[n - 1 - \ln \left(\frac{K}{S_t} \right) \right] P_t(S_t, K) dK \right\}, \text{ for } n \geq 2.\end{aligned}$$

The existence of a unique solution of system (2) for vector $\delta^{(m)}$ requires that matrix $\mathbf{Y}^{(m)}$ is invertible. The next proposition provides a general condition for this.

Proposition 1 *Let the m^{th} -order Hankel matrix of the sequence of the non-central risk neutral moments $\{\mu_{X,n}\}_{n=0}^{\infty}$, defined as $\mathbf{H}_{X,m} = \{\mu_{X,i+j-2}\}_{i,j=1}^m$, be positive definite, then matrix $\mathbf{Y}^{(m)}$ is invertible.*

The proof of the proposition is given in the Appendix.⁶ Given that, under the no arbitrage condition, the RND exists and $\{\mu_{X,n}\}_{n=0}^{\infty}$ defined by (5) constitutes the sequence of its moments, the Hankel matrix $\mathbf{H}_{X,m}$ will be positive definite by the converse of Hamburger's theorem and, hence, matrix $\mathbf{Y}^{(m)}$ will be invertible.⁷

III. Simulation Study

In this section we conduct a simulation study with the aim to assess if the C-GCSE can be successfully employed to approximate the RND of the log-return X from option prices based on the proposed method in the previous section. To this end, we compare our method to the one suggested by Corrado and Su (1996a, 1996b, 1997) -henceforth CS- based on the A-GCSE of the RND and a competing method which retrieves the RND by inverting the conditional characteristic function (CF) of X , denoted as $\psi_X(u)$, implied by option prices

⁶ $\mathbf{H}_{X,m} = \{\mu_{X,i+j-2}\}_{i,j=1}^m$ can be written as $\mathbf{H}_{X,m} = \begin{bmatrix} \mu_{X,0} & \mu_{X,1} & \cdots & \mu_{X,m-1} \\ \mu_{X,1} & \mu_{X,2} & & \mu_{X,m} \\ \vdots & & \cdots & \vdots \\ \mu_{X,m-1} & \mu_{X,m} & & \mu_{X,2m-2} \end{bmatrix}$.

⁷Hamburger's theorem states that a necessary and sufficient condition for the existence of a distribution with infinite support having moments $\{\mu_{X,n}\}_{n=0}^{\infty}$ is that the Hankel matrix $\{\mu_{X,i+j-2}\}_{i,j=1}^m$ is positive definite, $\forall m > 0$ (see Shohat and Tamarkin (1943), for a survey).

using the Inverse Fourier Transform (IFT).⁸ This CF can be derived in closed form based on Bakshi and Madan's (2000) fundamental option pricing theorem stating that $\psi_X(u)$ can be spanned by the OTM option prices. Its functional form is given by

$$(6) \quad \begin{aligned} \psi_X(u) = & 1 + iu(e^{r(T-t)} - 1) + e^{r(T-t)}iu(iu - 1) \left[\int_{S_t}^{+\infty} \frac{1}{K^2} e^{iu \ln(\frac{K}{S_t})} C_t(S_t, K) dK \right. \\ & \left. + \int_0^{S_t} \frac{1}{K^2} e^{iu \ln(\frac{K}{S_t})} P_t(S_t, K) dK \right], \end{aligned}$$

where i stands for the imaginary unit (see also Rompolis and Tzavalis (2004)).

In our simulation study, we assume that option prices are generated from the stochastic volatility (SV) option pricing model suggested by Heston (1993). This model considers that the risk neutralized stochastic processes for the stock price S_t and its volatility, V_t , are given as

$$(7) \quad \begin{aligned} dS_t &= S_t \left(rdt + \sqrt{V_t} dW_{t,1} \right), \text{ with} \\ dV_t &= \kappa(\vartheta - V_t)dt + \sigma \sqrt{V_t} dW_{t,2}, \end{aligned}$$

where $W_{t,1}$ and $W_{t,2}$ are two correlated Brownian motions with correlation coefficient ζ . It is frequently used in practice to improve upon the pricing performance of the BS model as it implies high levels of negative skewness and kurtosis of the RND. These two features of the RND are consistent with the pattern of the implied volatility, across different strike prices (see Bakshi, Cao and Chen (1997), and Chernov and Ghysels (2000)).

To generate empirically plausible option prices from model (7), we are based on values of its structural parameters found in the empirical literature (see Bakshi, Cao and Chen (1997)). These are set to the following levels: $\vartheta = 0.04$, $\sigma = 0.44$, $\kappa = 1.62$ and $\zeta = -0.76$. The strike prices K are taken to span the closed interval $[K_{\min} = 820, K_{\max} = 1260]$ at

⁸ Assuming a 4th-order A-GCSE of the RND, Corrado and Su (1996b) derived the following formulas for the call and put option prices:

$$C_t(S_t, K) = C_t^{BS}(S_t, K) + E_{X^*,3}Q_3 + E_{X^*,4}Q_4$$

and

$$P_t(S_t, K) = P_t^{BS}(S_t, K) + E_{X^*,3}Q_3 + E_{X^*,4}Q_4,$$

respectively, where $C_t^{BS}(S_t, K)$ and $P_t^{BS}(S_t, K)$ are the BS call and put prices, $Q_3 = \frac{1}{3!} S_t \sigma \sqrt{T-t} (2\sigma \sqrt{T-t} - d) \eta(d)$, $Q_4 = \frac{1}{4!} S_t \sigma \sqrt{T-t} (d^2 - 3d\sigma \sqrt{T-t} - 1)^2 \eta(d)$, and $d = \frac{\ln(\frac{S_t}{K}) + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma \sqrt{T-t}}$. These formulas can be fitted into option prices to estimate the risk neutral parameters of the A-GCSE of the RND collected in the vector $(\sigma_X^2, E_{X^*,3}, E_{X^*,4})$.

every 20 points, where $K_{\min}$ and $K_{\max}$ denote the lower and upper values of K . These values of $K_{\min}$ and $K_{\max}$ are the average levels of the minimum and maximum strike prices of a sample of short term option prices written on the S&P 500 index from January 1996 to December 2002, which is used in our empirical study. To be consistent with the magnitude of them, the values of the two state variables S_t and V_t are taken to be the mean of the S&P500 index and the variance of its log-return over the sample period. These are given as $S_t = 1080$ and $V_t = 0.026$, respectively. For the interest rate r , we used the average level of the three-month US Treasury bill of the above period, given by $r = 0.05$.

Based on the above values of the SV option pricing model, we generated a set of $N = 22$ call option prices of a short-term to maturity interval $T - t = 2.5$ months (0.21 years), where the biggest failures of the BS model are observed. From this set, next we produced another one consisting of 22 put option prices using the put-call arbitrage relationship, thus bringing the total number of generated option prices to 44. This should be considered as a small cross-sectional sample of option prices used to estimate the RND, as those often met in practice. Following Ait-Sahalia and Duarte (2003), we added a noise term to our generated set of option prices drawn from the uniform distribution, with interval $[-0.01, 0.01]$. This term can be taken to reflect random effects of the bid-ask spread and the different degree of liquidity on option call/put prices.

In Table 1, we report the first set of our Monte Carlo exercise results based on a number of 1000 iterations. This table presents the average values of the first four well known descriptive statistics of the RND, i.e. the mean, variance and the skewness and kurtosis coefficients, over all iterations. These are derived by the following two methods. The first (denoted as RNM-C) gives estimates of these statistics based on values of the non-central moments $\mu_{X,j}$ implied by option prices using the RNM formulas (5). The second (denoted as RNM-A) provides estimates of them based on the procedure suggested by the CS method, based on the A-GCSE (see fn 8). This is done by imposing the nonnegative constraint $f_X(x) \geq 0$ in the estimation of the RND, as often done in practice. Comparing the performance of the above two methods is crucial for understanding the reasons why our approach, employing the C-GCSE, can outperform that based on the A-GCSE in providing accurate estimates of the four distribution statistics and the RND itself. To assess this, the table reports the theoretical values of the statistics implied by the SV model and the values of two well known error valuation metrics: the mean absolute error (MAE) and the

mean percentage error (MPE) between the theoretical and estimated values by the above two methods considered, over the 1000 iterations.

The results of Table 1 clearly show that the RNM-C method can provide very accurate estimates of the basic distribution statistics examined. The MAE and MPE metrics take values close to zero for all the statistics, while the standard deviations reported in parentheses imply that their average estimates are not different from their predicted values, at 5% significance level. On the other hand, the results of the table indicate that the estimates of the four statistics derived by the RNM-A method substantially differ from their theoretical values, especially the skewness and kurtosis coefficients. The first of these coefficients is biased downwards, and it has a very high MPE (67.72%). This is obviously due to the density constraint $f_X(x) \geq 0$ employed in the implementation of the RNM-A method. Note that the theoretical value of it reported in the table indicates that, if the above constraint was not imposed in estimation, then its value would lie outside the interval $[-1.0493, 1.0493]$ implying negative estimates of the RND. Apart from the skewness coefficient, the above constraint can also lead to a downward bias of the estimate of the variance σ_X^2 and, hence, the risk neutral mean, given by $r(T - t) - \frac{1}{2}\sigma_X^2$ (see fn 3). Although the magnitude of this bias appears to be very small in our simulation study (i.e. it has a MPE which is 2.7668% for σ_X^2 and 0.1122% for the mean), in practice it can be very high as shown in our empirical exercise presented in the next section.

Given that an optimal order of the C-GCSE of the RND is chosen, the ability of the RNM-C method to retrieve accurate estimates of the risk neutral statistics should imply that our method can provide efficient estimates of the RND over its whole interval support. To examine this more formally, next we report the values of two well known metrics of density accuracy for our method (denoted as $\text{RND}^{\text{C-GCSE}}$), over the 1000 Monte Carlo iterations. These are the L^1 and L^2 metrics, defined as

$$L^1 = \int |f_X(x) - f_X^*(x)| dx \quad \text{and} \quad L^2 = \int (f_X(x) - f_X^*(x))^2 dx,$$

respectively, where $f_X^*(x)$ denotes the true probability density function. To compare the performance of our method with the two others, these metrics were also calculated for the CS method (denoted as $\text{RND}^{\text{A-GCSE}}$) and that recovering the RND by inverting the CF (6)

through the IFT (denoted as RND^{ICF}).⁹ The theoretical (true) values of the RND which are used to calculate the values of L^1 and L^2 are obtained by inverting the theoretical CF implied by the SV option pricing model. In implementing our method, note that we set the optimal order of the C-GCSE to $m = 4$ based on the SC information criterion.

The reported values of the metrics L^1 and L^2 are given in the following table:

	$\text{RND}^{\text{A-GCSE}}$	$\text{RND}^{\text{C-GCSE}}$	RND^{ICF}
L^1	0.3169	0.0781	0.1689
L^2	0.3587	0.0153	0.1049

These values confirm the superiority of our method over the two others considered, especially the $\text{RND}^{\text{A-GCSE}}$. This conclusion is also supported by inspecting Figure 1, which graphically presents the estimates of the RND derived by the three alternative methods against the true density. The superiority of our method compared with the $\text{RND}^{\text{A-GCSE}}$ is obviously due to the fact that it relies on more accurate estimates of the risk neutral moments employed in this type expansion of the RND. It also due to its ability to account for stronger deviations from normality than the $\text{RND}^{\text{A-GCSE}}$ method. On the other hand, its better performance than the RND^{ICF} method can be attributed to the numerical error encountered when approximating the continuous integrals involved in CF (6) based on a discrete set of option prices. As more formally is shown in the Appendix, the magnitude of this error has a bound which increases with u due to the trigonometric nature of function (6). This function is difficult to span with a finite number of options. When applying the IFT to invert function (6), this error becomes even bigger due to the integration over absolute values of this function involved in this transformation. As Figure 1 reveals, it can cause considerable oscillation in the estimates of the RND and it can also result in negative probabilities.

IV. Empirical Exercise

In this section, we conduct an empirical exercise with the aim of examining if there are any price valuation (prediction) gains in employing our method of approximating the RND

⁹To estimate the CF (6) from a discrete set of option prices over different strike prices K_i , we have followed the same procedure as the one employed for the risk neutral statistics, reported in Table 1. That is, we have employed cubic splines to interpolate the implied volatilities between two different points, while we use linear extrapolation for the strike intervals $(0, K_{\min}]$ and $[K_{\max}, +\infty)$.

based on the C-GCSE (i.e. $\text{RND}^{\text{C-GCSE}}$) to provide out-of-sample-forecasts of market option prices.¹⁰ To this end, we compare our method with that based on the $\text{RND}^{\text{A-GCSE}}$, often used in practice to predict future option prices (see Corrado and Su (1997)).¹¹ In particular, our exercise can show if by employing the $\text{RND}^{\text{C-GCSE}}$ method of approximating the true RND in option pricing can reduce the valuation errors. In our analysis, we use a sample of European call and put option prices written on the S&P500 index. The maturity interval of our options is close to 22 trading days. Our sample covers the period from January 1996 to December 2002.

To study the distributional features of the RND implied by our data, our analysis starts with presenting descriptive statistics of the RND and its estimates of it over the whole sample period. Table 2 reports the average estimates of the values of the first four distribution statistics obtained by the RNM-C and RNM-A methods at each point in time of our sample, while Figure 2 presents the estimates of the RND based on the $\text{RND}^{\text{C-GCSE}}$ and $\text{RND}^{\text{A-GCSE}}$ methods, respectively. This is a crucial step in our analysis as it can indicate potential sources of option pricing prediction failures of the above two methods, across different moneyness levels. As was expected, the results of the table and figure reveal that our set of option prices imply a RND with high degree of kurtosis and negative skewness. The estimates of the skewness and kurtosis coefficients obtained by the RNM-C method appear to lie far outside the bounds required by the A-GCSE in order for it to provide positive risk neutral probabilities.¹² In addition to these two statistics, Table 2 also shows that there is a significant difference between the estimate of the risk neutral mean obtained by the two methods. The estimate provided by the RNM-C method is significantly bigger than that given by the RNM-A. This result explains the different location of the two estimates of the RND, shown in Figure 2.

The above differences in the estimates the RND and its descriptive statistics can be translated into significant valuation gains in employing the $\text{RND}^{\text{C-GCSE}}$ method of esti-

¹⁰This is a standard exercise carried out in the literature to evaluate the empirical performance of many option pricing models or their implied risk neutral densities by option prices (see Bakshi, Cao and Chen (1997), and Heston and Nandi (2000)).

¹¹We are focused on comparing the C-GCSE with the A-GCSE, as we have found that the ICF method provides negative estimates of the risk neutral probabilities for our option prices data set.

¹²Our estimate of the skewness coefficient is close to that found by Dennis and Mayhew (2002) based on a set of S&P500 option prices of the same maturity covering the period from April 7, 1986, to December 31, 1996.

imating the RND to predict future option prices, compared with the $\text{RND}^{\text{A-GCSE}}$. As was stated before, this will happen when the $\text{RND}^{\text{C-GCSE}}$ method approximates better than the $\text{RND}^{\text{A-GCSE}}$ the true RND of the data. To investigate this more formally, in Table 3 we present performance evaluation measures of our out-of-sample forecasting exercise. This was set up as follows. On the third Wednesday of each month, we retrieved estimates of the RND based on the $\text{RND}^{\text{C-GCSE}}$ and $\text{RND}^{\text{A-GCSE}}$ methods, respectively, using our set of option prices at each point in time over the whole sample period. These estimates were then used to predict one-period ahead prices of call and put options (i.e. for the third Thursday of each month), with one period less to maturity horizon. For our method, the predicted values of the option prices were derived by numerically evaluating the standard Cox-Ross option pricing formula using estimates of the RND based on the $\text{RND}^{\text{C-GCSE}}$ method. For the method based on the $\text{RND}^{\text{A-GCSE}}$, we used the option pricing formulas suggested by Corrado and Su (1996b) [see fn 8]. The values of the evaluation measures that the table reports are the MAE and MPE between the predicted and the actual option prices, over the whole sample period. In addition to these measures, the table also reports values of Theil's inequality coefficient (see Greene (2003)), calculated over our whole set of data as

$$U = \sqrt{\frac{\sum_{t=1}^T \sum_{i=1}^N \left\{ [\Delta C_t^{\text{obs}}(S_t, K_i) - \Delta C_t(S_t, K_i)]^2 + [\Delta P_t^{\text{obs}}(S_t, K_i) - \Delta P_t(S_t, K_i)]^2 \right\}}{\sum_{t=1}^T \sum_{i=1}^N [\Delta C_t^{\text{obs}}(S_t, K_i) + \Delta P_t^{\text{obs}}(S_t, K_i)]^2}},$$

where Δ stands for the difference operator, and $C_t^{\text{obs}}(S_t, K_i)$ and $P_t^{\text{obs}}(S_t, K_i)$ denote the observed values of call and put prices, respectively, over the cross-sectional ($i = 1, 2, \dots, N$) and time-series ($t = 1, 2, \dots, T$) dimensions of our sample. This coefficient can assess the ability of the above two methods to satisfactorily forecast turning points (i.e. the correct sign changes) of future option prices, which is generally considered as a difficult econometric task. These turning points can reflect changes in market conditions about the future price of the underlying asset and its implied RND due for instance to shifts from the bull to the bear market regime. The smaller the magnitude of Theil's coefficient, the better the forecasting performance of our method will be.

The results of the table clearly indicate that our approach, based on the $\text{RND}^{\text{C-GCSE}}$ method, considerably improves upon the performance of the method based on the $\text{RND}^{\text{A-GCSE}}$ to predict future option prices or their changes. This can be supported by all forecasting

evaluation metrics employed and it is true across the entire implied volatility curve (i.e. OTM puts, ATM calls and puts and OTM calls). The highest benefits of our method are observed for the OTM call and put options. Obviously, the better performance of the $\text{RND}^{\text{C-GCSE}}$ method reflects that fact that it approximates more accurately the correct location and the tails of the true RND than the $\text{RND}^{\text{A-GCSE}}$. The small values of the Theil's coefficient reveal that our method is also robust to possible changes in market conditions through the sample period.

V. Conclusions

In this paper we have suggested a new method of retrieving the RND of future stock, or any other asset, returns implied by European call and put prices based on an exponential form of Gram-Charlier series expansion (GCSE), known as C-type. This type of GCSE can encompass the main drawbacks of the standard GCSE (known as A-type), namely to provide negative probabilities or not accounting for stronger deviations from the normal distribution. The implementation of this expansion to recover the RND only requires estimates of its risk neutral moments. These can retrieve from option prices in a model-free manner, following Bakshi and Madan's (2000) methodology. Thus, our method can be employed to estimate the RND even for option pricing models whose the implied characteristic function or RND are not given in closed form solution.

To assess the performance the method, the paper conducts a simulation study generating option prices from the stochastic volatility model. It also carries out an empirical exercise investigating if there are any price valuation gains in employing the method to predict option prices, compared to a method based on the A-type GCSE often used in practice. The results of both of these exercises support the use of the C-type GCSE in option pricing. They clearly show that it can be considerably improve upon the performance of the A-type GCSE of the RND.

VI. Appendix

In the Appendix we provide a proof of Proposition 1 and derive the bounds of the numerical approximation error of the CF $\psi_X(u)$, given by (6), and its associated RND incurred when implementing the ICF method to retrieve the RND.

Proof of Proposition 1: First note that the ij -element of matrix $\mathbf{Y}^{(m)}$ is given by:

$$Y_{ij}^{(m)} = \sum_{n=0}^{i+j-2} \frac{1}{n!} \Delta_{i-1,j-1,n} E_{X^*,n}.$$

Based on definition of Hermite polynomials

$$H_n(x^*) = \sum_{l=0}^{\lfloor n/2 \rfloor} (-1)^l \frac{n!}{2^l l!} x^{*n-2l}$$

and the general result

$$\int \sum_{n=0}^{i+j-2} \frac{1}{n!} \Delta_{i-1,j-1,n} H_n(x^*) f_{X^*}(x^*) dx^* = \int H_{i-1}(x^*) H_{j-1}(x^*) f_{X^*}(x^*) dx^*,$$

which holds for them (see Muscolino and Ricciardi (1999)), it can be shown that the elements of matrix $\mathbf{Y}^{(m)}$ can be written as

$$Y_{ij}^{(m)} = \sum_{k=0}^{\lfloor (i-1)/2 \rfloor} \sum_{l=0}^{\lfloor (j-1)/2 \rfloor} (-1)^k \frac{(i-1)!}{2^k k!} (-1)^l \frac{(j-1)!}{2^l l!} \mu_{X^*, i+j-2-2(l+k)},$$

or, after rearranging terms, as

$$(8) \quad Y_{ij}^{(m)} = \sum_{k=0}^{i-1} \sum_{l=0}^{j-1} a_{ik+1} a_{jl+1} \mu_{X^*, k+l},$$

where

$$a_{ij} = (-1)^{(i-j)/2} \frac{(i-1)!}{2^{(i-j)/2} [(i-j)/2]!} \mathbf{1}_{\{i+j=\text{even}\}},$$

$$\text{with } \mathbf{1}_{\{i+j=\text{even}\}} = \begin{cases} 1 & \text{if } i+j = \text{even} \\ 0 & \text{elsewhere} \end{cases}.$$

Equation (8) implies that matrix $\mathbf{Y}^{(m)}$ can be decomposed as

$$(9) \quad \mathbf{Y}^{(m)} = \mathbf{A} \mathbf{H}_{X^*,m} \mathbf{A}',$$

where $\mathbf{H}_{X^*,m} = \{\mu_{X^*,i+j-2}\}_{i,j=1}^m$ and $\mathbf{A}$ is a $(m \times m)$ lower triangular matrix with elements:

$$A_{ij} = \begin{cases} 0 & \text{if } i < j \\ a_{ij} & \text{if } i \geq j \end{cases}.$$

The determinant of $\mathbf{A}$ is $\det(\mathbf{A}) = \det(\mathbf{A}') = 1$, since $A_{ii} = 1 \forall i = 1, 2, \dots, m$.

Using (4), the ij -element of the matrix $\mathbf{H}_{X^*,m}$ can be written as

$$\mu_{X^*,i+j-2} = \frac{1}{\sigma_X^{i+j-2}} \sum_{h=0}^{i+j-2} \binom{i+j-2}{h} (-\mu_{X,1})^h \mu_{X,i+j-2-h}.$$

By the Vandermonde convolution formula [see Riordan (1979)], the last formula of $\mu_{X^*,i+j-2}$ after rearranging terms can be written as

$$(10) \quad \mu_{X^*,i+j-2} = \sum_{k=0}^{i-1} \sum_{l=0}^{j-1} b_{ik+1} b_{jl+1} \mu_{X,k+l},$$

where

$$b_{ij} = \frac{1}{\sigma_X^{i-1}} \binom{i-1}{i-j} (-\mu_{X,1})^{i-j}.$$

Equation (10) implies that matrix $\mathbf{H}_{X^*,m}$ can be decomposed as follows

$$(11) \quad \mathbf{H}_{X^*,m} = \mathbf{B} \mathbf{H}_{X,m} \mathbf{B}',$$

where $\mathbf{H}_{X,m}$ is positive definite by the condition of the proposition and $\mathbf{B}$ is a $(m \times m)$ lower triangular matrix with elements given as

$$B_{ij} = \begin{cases} 0 & \text{if } i < j \\ b_{ij} & \text{if } i \geq j \end{cases}.$$

The determinant of $\mathbf{B}$ is $\det(\mathbf{B}) = \det(\mathbf{B}') = \frac{1}{\sigma_X^{m(m-1)/2}} > 0$, since $B_{ii} = \frac{1}{\sigma_X^{i-1}} \forall i = 1, 2, \dots, m$.

Substituting equation (11) into (9) yields

$$\mathbf{Y}^{(m)} = \mathbf{A} \mathbf{B} \mathbf{H}_{X,m} \mathbf{B}' \mathbf{A}'$$

which implies that $\mathbf{Y}^{(m)}$ is positive definite, as $\mathbf{H}_{X,m}$ is positive definite and $\det(\mathbf{A} \mathbf{B}) =$

$\det(\mathbf{B}'\mathbf{A}') > 0$. This proves that $\det(\mathbf{Y}^{(m)}) > 0$ and, hence, the result of Proposition 1 that $\mathbf{Y}^{(m)}$ is invertible.

Bounds of the Numerical Approximation Errors of the ICF Method

Let $\hat{\psi}_X(u)$ denote the estimate of the CF $\psi_X(u)$, given by (6), based on a discrete set of OTM option prices. Then, we can show that the numerical errors of the real (Re) and imaginary (Im) parts of $\psi_X(u)$ encountered when approximating the continuous integrals involved in (6) between discrete points of the data are bounded as follows:

$$(12) \quad \left| \operatorname{Re} \psi_X(u) - \operatorname{Re} \hat{\psi}_X(u) \right| \leq A \cdot F_{\operatorname{Re}}(u) \quad \text{and} \quad \left| \operatorname{Im} \psi_X(u) - \operatorname{Im} \hat{\psi}_X(u) \right| \leq A \cdot F_{\operatorname{Im}}(u)$$

respectively, where

$$A = \int_{S_t}^{K_{\max}} \left| C_t(S_t, K) - \hat{C}_t(S_t, K) \right| dK + \int_{K_{\min}}^{S_t} \left| P_t(S_t, K) - \hat{P}_t(S_t, K) \right| dK$$

is the approximation error of the option pricing function, where $\hat{C}_t(S_t, K)$ and $\hat{P}_t(S_t, K)$ denote the approximated values of the call and put option prices, respectively, and quantities $F_{\operatorname{Re}}(u)$ and $F_{\operatorname{Im}}(u)$ are defined as

$$F_{\operatorname{Re}}(u) = \max_{K \in (K_{\min}, K_{\max})} e^{r(T-t)} \left| u \frac{1}{K^2} \sin \left[u \ln \left(\frac{K}{S_t} \right) \right] - u^2 \frac{1}{K^2} \cos \left[u \ln \left(\frac{K}{S_t} \right) \right] \right|$$

and

$$F_{\operatorname{Im}}(u) = \max_{K \in (K_{\min}, K_{\max})} e^{r(T-t)} \left| u^2 \frac{1}{K^2} \sin \left[u \ln \left(\frac{K}{S_t} \right) \right] + u \frac{1}{K^2} \cos \left[u \ln \left(\frac{K}{S_t} \right) \right] \right|,$$

respectively.

The bounds given by equation (12) show that the numerical error incurred when approximating the CF $\psi_X(u)$ with the above numerical methods critically depends on the sine (sin) and cosine (cos) trigonometric functions involved in its real and imaginary parts. These determine the amplitude of the oscillation of $\hat{\psi}_X(u)$ which increases with u . By applying the IFT to obtain the RND from $\hat{\psi}_X(u)$, denoted as $\hat{f}_X(u)$, this numerical approximation error will increase in magnitude because the IFT integrates $\hat{\psi}_X(u)$ over the absolute values of its real and imaginary parts. It can be shown that the numerical approximation error of

$\hat{f}_X(u)$ is bounded as

$$\left| f_X(x) - \hat{f}_X(x) \right| \leq E,$$

where $E = \frac{1}{\pi} \int_0^A \left[\left| \operatorname{Re} \psi_X(u) - \operatorname{Re} \hat{\psi}_X(u) \right| + \left| \operatorname{Im} \psi_X(u) - \operatorname{Im} \hat{\psi}_X(u) \right| \right] du$.

References

- Abadir, K. M. and M. Rockinger. "Density Functionals, with an Option-Pricing Application.", *Econometric Theory*, 19 (2003), 778-811.
- Ait-Sahalia, Y. "Maximum Likelihood Estimation of Discretely Sampled Diffusion: A Closed-Form Approximation Approach." *Econometrica*, 70 (2002), 223-262.
- Ait-Sahalia, Y. and A. W. Lo. "Nonparametric Estimation of State-Price Densities Implicit in Financial Asset Prices." *Journal of Finance*, 53 (1998), 499-548.
- Ait-Sahalia, Y. and J. Duarte. "Nonparametric Option Pricing under Shape Restrictions." *Journal of Econometrics*, 116 (2003), 9-47.
- Bakshi, G., C. Cao and Z. Chen. "Empirical Performance of Alternative Option Pricing Models." *Journal of Finance*, 52 (1997), 2003-2049.
- Bakshi, G., N. Ju and H. Ou-Yang. "Estimation of Continuous-Time Models with an Application to Equity Volatility Dynamics." *Journal of Financial Economics*, 82 (2006), 227-249.
- Bakshi, G., N. Kapadia and D. Madan. "Stock Return Characteristics, Skew Laws, and the Differential Pricing of Individual Equity Options." *Review of Financial Studies*, 16 (2003), 101-143.
- Bakshi, G. and D. Madan. "Spanning and Derivative Security Valuation." *Journal of Financial Economics*, 55 (2000), 205-238.
- Bliss, R. and N. Panigirtzoglou. "Testing the Stability of Implied Probability Density Functions." *Journal of Banking and Finance*, 26 (2002), 381-422.
- Campa, J. M., P. H. K. Chang and R. L. Reider. "Implied Exchange Rate Distributions: Evidence from OTC Option Markets." *Journal of International Money and Finance*, 17 (1998), 117-160.
- Charlier, C.V.L. "A New Form of the Frequency Function." *Maddalende fran Lunds Astronomiska Observatorium*, Ser II (1928), 51.
- Chernov, M. and E. Ghysels. "A Study Towards a Unified Approach to the Joint Estimation of Objective and Risk Neutral Measures for the Purpose of Options Valuation." *Journal of Financial Economics*, 56 (2000), 407-458.

- Corrado, C. and T. Su. "S&P 500 Index Option Tests of Jarrow and Rudd's Approximate Option Valuation Formula." *Journal of Futures Markets*, 16 (1996a), 611-629.
- Corrado, C. and T. Su. "Skewness and Kurtosis in S&P 500 Index Returns Implied by Option Prices." *Journal of Financial Research*, 19 (1996b), 175-192.
- Corrado, C. and T. Su. "Implied Volatility Skews and Stock Index Skewness and Kurtosis Implied by S&P 500 Index Option Prices." *Journal of Derivatives*, 4 (1997), 8-19.
- Cox, J. and S. Ross. "The Valuation of Options for Alternative Stochastic Processes." *Journal of Financial Economics*, 3 (1976), 145-166.
- Dennis, P. and S. Mayhew. "Risk Neutral Skewness: Evidence from Option Prices." *Journal of Financial and Quantitative Analysis*, 37 (2002), 471-493.
- Flamouris, D. and D. Giamouridis. "Estimating Implied PDF's from American Options and Futures: A New Semiparametric Approach." *Journal of Futures Markets*, 22 (2002), 1-30.
- Greene, W. "Econometric Analysis." Prentice Hall International Edition (2003).
- Heston, S. L. "A Closed-Form Solution for Options with Stochastic Volatility with Applications to Bond and Currency Options." *Review of Financial Studies*, 6 (1993), 327-343.
- Heston, S. L. and S. Nandi. "A Closed-Form GARCH Option Valuation Model." *Review of Financial Studies*, 13 (2000), 585-625.
- Jackwerth, J. C. and M. Rubinstein. "Recovering Probability Distributions from Options Prices." *Journal of Finance*, 51 (1996), 1611-1631.
- Jarrow, R. and A. Rudd. "Approximate Valuation for Arbitrary Stochastic Processes." *Journal of Financial Economics*, 10 (1982), 347-369.
- Jondeau, E. and M. Rockinger. "Gram-Charlier Densities." *Journal of Economic Dynamics and Control*, 25 (2001), 1457-1483.
- Kendall, M. G. and A. Stuart. "The Advanced Theory of Statistics." Hafner Publishing Company, New York (1969).

- Knight, J. and S. Satchell. "Pricing Derivatives Written on Assets with Arbitrary Skewness and Kurtosis." In *Return Distributions in finance*, Knight and Satchell (eds.), Butterworth and Heinemann (2001), 252-275.
- Longstaff, F. A. "Option Pricing and the Martingale Restriction." *Review of Financial Studies*, 8 (1995), 1091-1124.
- Madan, D. B. and F. Milne. "Contingent Claims Valued and Hedged by Pricing and Investing in a Basis." *Mathematical Finance*, 4 (1994), 223-245.
- Melick, W. R. and C. P. Thomas. "Recovering an Asset Implied PDF from Option Prices: An Application to Crude Oil During the Gulf Crisis." *Journal of Financial and Quantitative Analysis*, 32 (1997), 91-115.
- Muscolino, G. and G. Ricciardi. "Probability Density Function of MDOF Structural Systems Under Non-Normal Delta-Correlated Inputs." *Comput. Methods Appl. Mech. Engrg.*, 168 (1999), 121-133.
- Pagan, A. and A. Ullah. "Nonparametric Econometrics." Cambridge University Press (1999).
- Riordan, J. "Combinatorial Identities." Robert E. Krieger Publishing Co, New York (1979).
- Rompolis, L. S. and E. Tzavalis. "Retrieving Risk Neutral Moments from Option Prices." Mimeo, Department of Accounting and Finance, Athens University of Economics and Business (2004).
- Rubinstein, M. "Implied Binomial Trees." *Journal of Finance*, 49 (1994), 771-818.
- Shohat, J. A. and J. D. Tamarkin. "The Problem of Moments." Providence, RI: AMS Mathematical Survey (1943).

TABLE 1

Descriptive Statistics and Measurement Errors of the Simulation Study

Theoretical		RNM-C method			RNM-A method		
			MAE	MPE		MAE	MPE
Mean	0.0075455	0.0075455 (1.96×10^{-5})	4.59×10^{-6}	0.06%	0.0075363 (2.03×10^{-5})	8.46×10^{-6}	0.11%
Variance	0.006093	0.006094 (3.88×10^{-5})	9.82×10^{-6}	0.16%	0.005927 (4.06×10^{-5})	0.00016	2.77%
Skewness	-1.21101	-1.21276 (0.03218)	0.007679	0.63%	-0.39706 (0.00667)	0.81681	67.43%
Kurtosis	5.46637	5.48414 (0.12798)	0.076748	1.40%	5.95131 (0.19898)	0.50220	9.21%

Notes: (i) This table presents the mean values over 1000 iterations of the first four distribution statistics, i.e. the mean, variance, skewness and kurtosis coefficients of the RND implied by the SV model, with standard deviations in parentheses. It also presents the mean absolute error (MAE) and the mean percentage error (MPE) between the theoretical and the estimated values of the above statistics. The theoretical values of the four descriptive statistics are derived based on the CF of the SV model derived by Heston (1993), given as $\phi_X(u) = e^{ru(T-t)+L(u)+M(u)V_t}$, where $L(u) = \frac{\kappa\theta}{\sigma^2}[(\kappa-\zeta\sigma ui+d)(T-t)-2\ln\left(\frac{1-qe^{d(T-t)}}{1-q}\right)]$, $M(u) = \frac{\kappa-\zeta\sigma ui+d}{\sigma^2}\left(\frac{1-qe^{d(T-t)}}{1-q}\right)$, with $q = \frac{\kappa-\zeta\sigma ui+d}{\kappa-\zeta\sigma ui-d}$ and $d = \sqrt{(\zeta\sigma ui - \kappa)^2 - \sigma^2 ui(ui-1)}$. From this CF, we derived the theoretical values of the j^{th} -order non-central moments of the RND by numerically calculating the j^{th} -order derivative of $\phi_X(u)$ at $u = 0$, i.e. $\mu_{X,j} = \frac{1}{i^j} \left[\frac{d^j \phi_X(u)}{du^j} \right]_{u=0}$. These values of $\mu_{X,j}$ are then used to calculate the theoretical values of the descriptive statistics of the table. (ii) The RNM-C method gives values of the statistics directly from option prices based on the RNM formulas (5). As these formulas employ integrals of continuous functions to retrieve the values of the risk neutral moments, we have employed cubic splines to interpolate the implied by our option prices volatilities between two different points of the data (see Campa et al (1998)). Due to the lack of option prices at 0 and $+\infty$, we have extrapolated the implied volatilities by a linear function over the intervals $(0, K_{\min}]$ and $[K_{\max}, +\infty)$. (iii) The RNM-A method gives estimates of the risk neutral moments based on the CS method (see fn 8). Based on the non-linear least squares method, this method fits the option pricing formulas for European calls/puts derived by Corrado and Su (1996b) into option prices.

TABLE 2
Descriptive Statistics

	RNM-C method	RNM-A method	t-statistic
Mean	0.0018 (0.00213)	0.00048 (0.00339)	5.82
Variance	0.00493 (0.00263)	0.00485 (0.00269)	0.97
Skewness	-2.0141 (0.5789)	-0.49045 (0.12328)	-24.92
Kurtosis	15.4093 (8.0359)	6.22757 (0.74359)	10.48

Notes: This table presents the average values of the estimates of the first four descriptive statistics retrieved at each point in time of the whole sample period of our data set. Standard deviations are in parentheses. The fourth column of the table gives the values of the t-statistic for the difference in means between the values of the statistics derived by the RNM-C and RNM-A methods.

TABLE 3
Out-of-Sample Option Pricing Forecasting Performance

	OTM puts	ATM puts	All puts	ATM calls	OTM calls	All calls	All options
	MAE						
A-GCSE	1.37	6.20	3.95	5.17	2.63	5.31	4.58
C-GCSE	1.23	5.10	3.19	5.11	1.25	4.78	3.91
	MPE%						
A-GCSE	42.88	24.01	33.41	21.20	201.13	85.03	56.54
C-GCSE	38.97	20.30	30.30	19.56	36.23	22.94	26.92
	Theil's coefficient (U)						
A-GCSE	0.8200	1.2290	0.2017	1.3521	1.6486	0.3084	0.3833
C-GCSE	0.6970	0.9775	0.1467	1.3078	0.8617	0.2832	0.3157

Notes: This table presents the MAE and MPE valuation metrics for our empirical exercise forecasting out-of-sample option prices. This is done across different moneyness levels defined as follows. A call (put) option is said to be at-the-money (ATM) if $S_t/K \in (0.97, 1.03)$ and out-of-the-money (OTM) if its $S_t/K \leq 0.97$ ($S_t/K \geq 1.03$). In addition to the above two metrics, the table presents Theil's inequality coefficient U .

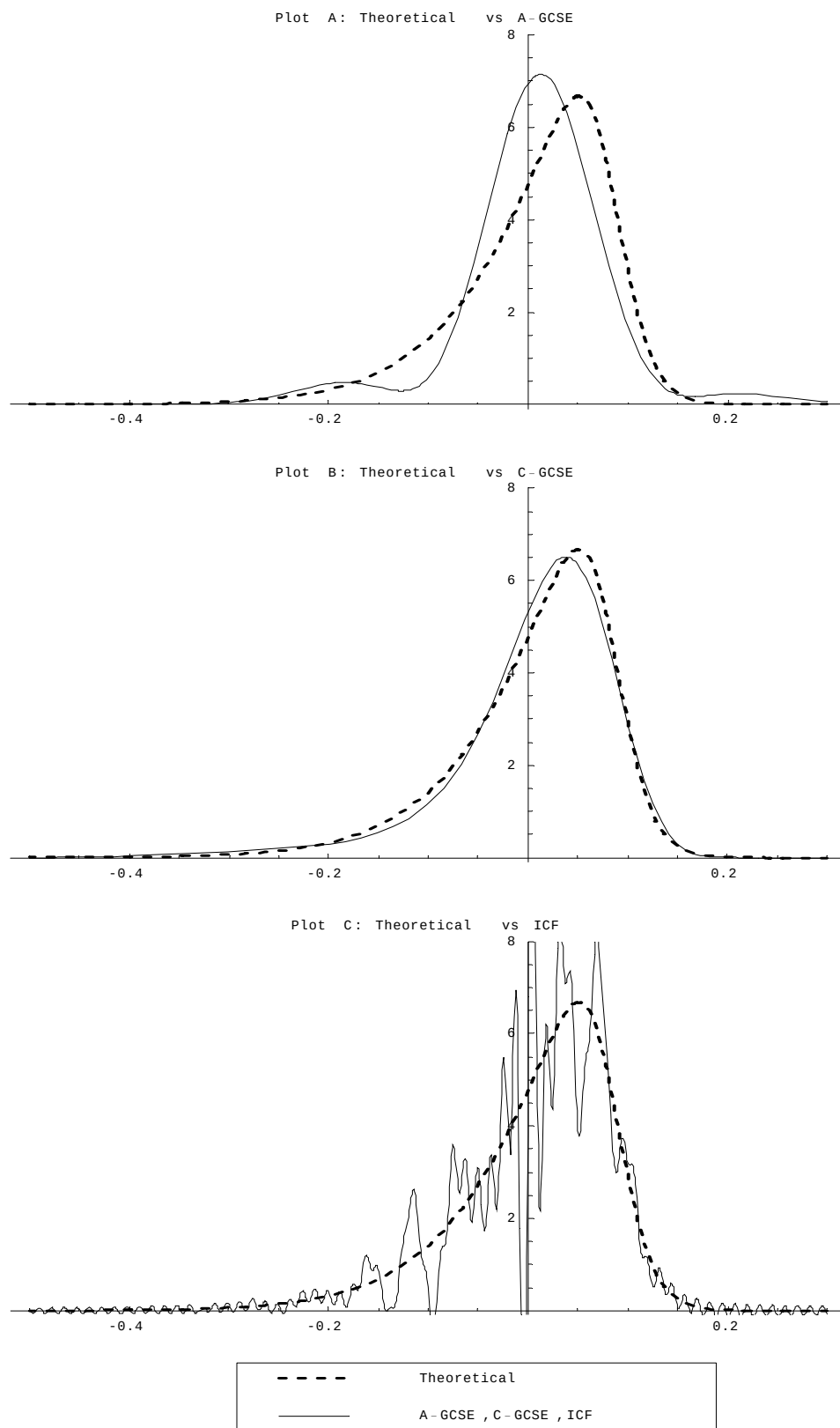


Figure 1: Estimates of the RND Implied by the SV Model.

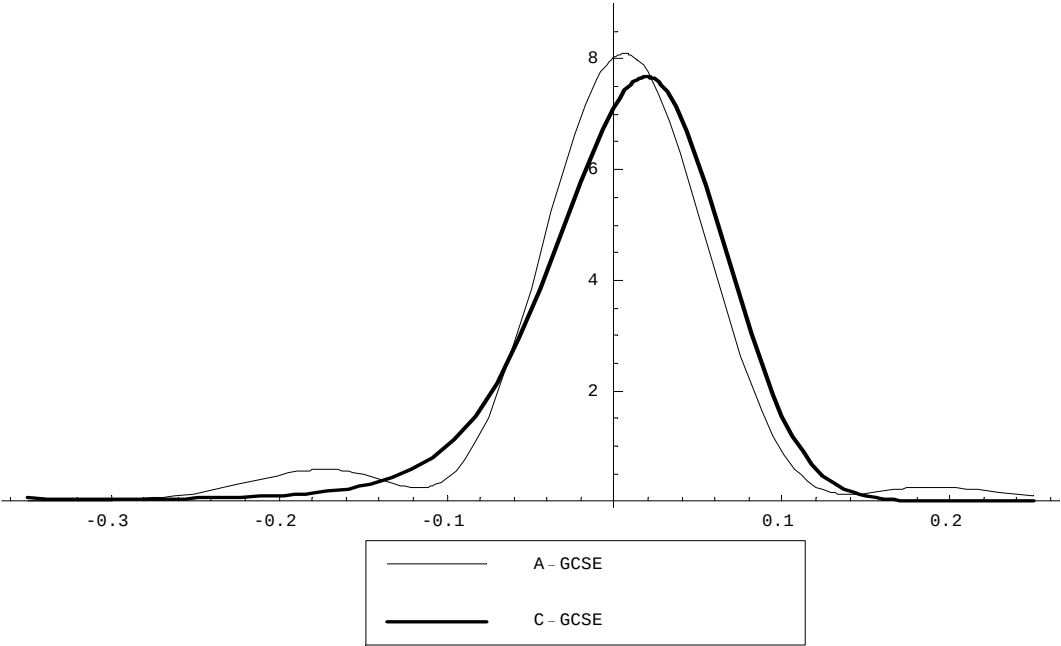


Figure 2: Estimates of the Approximating RND Based on the $\text{RND}^{\text{C-GCSE}}$ and $\text{RND}^{\text{A-GCSE}}$ Methods